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Studierichting: Informatica
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$$I = \int \frac{x^2 dx}{4x^2 + 1}$$

$$\begin{array}{cc|cc}
x^2 & & 4x^2 & +1 \\ \cline{3-4}
-x^2 & -\frac{1}{4} & \frac{1}{4} & \\ \cline{1-2}
0 & -\frac{1}{4} & &
\end{array}$$

$$\int \frac{1}{4} dx + \int \frac{\frac{-1}{4}}{4x^2 + 1} dx = \frac{1}{4} x - \frac{1}{4} \int \frac{1}{4x^2 + 1} dx$$

Stel: $t = 2x \Rightarrow dt = 2dx \Leftrightarrow dx = \dfrac{dt}{2}$

$$= \frac{1}{4} x - \frac{1}{4} \int \frac{1}{t^2 + 1} \frac{dt}{2} = \frac{1}{4} x - \frac{1}{8} \operatorname{Bgtan}(2x) + c$$

References